GATE
CS & DA

Artificial Intelligence

Miscellaneous

Lecture No. 2

By- SIDDHARTH SABHARWAL SIR

Recap of Previous Lecture

Topic Bayesian N/W

Topic

Topic

Topic

Topic

Turn on Slide map

Topics to be Covered

- Topic — Bayesian n/w
- Topic — Markov cycle
- Topic
- Topic
- Topic

In order to
SUCCEED
we must first believe than we can
Nikos Kazantzakis | www.geckoandfly.com

Basics of probability

- Independent RVs
- $P(AB) = P(A/B)P(B) = P(B/A)P(A) = P(A).P(B)$
- $P(A/B) = P(A)$
- $P(B/A) = P(B)$

Basics of probability

- Normal process to find
- P(ABC) $\Rightarrow P[C/AB]P(AB)$

  $\Rightarrow P[C/AB]P(B/A)P(A)$

- Simplex structure

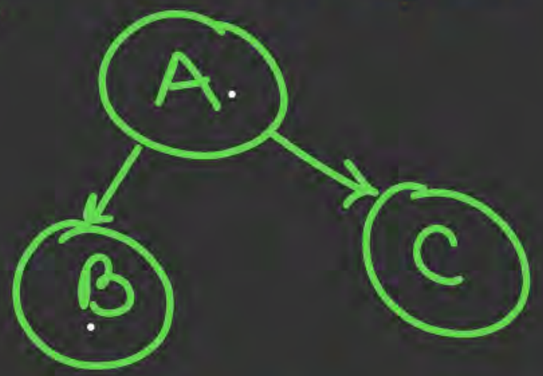
- $P(A)$ i.e $P(A=True)$ (directly)
- $P(B)$ i.e $P(B=True)$ ~~directly~~

$$\Downarrow$$

$$P(B/A)P(A)+P(B|\sim A)P(\sim A)$$

$$\rightarrow P(B=T, A=T)+P(B=T, A=F)$$

A is indepent of all



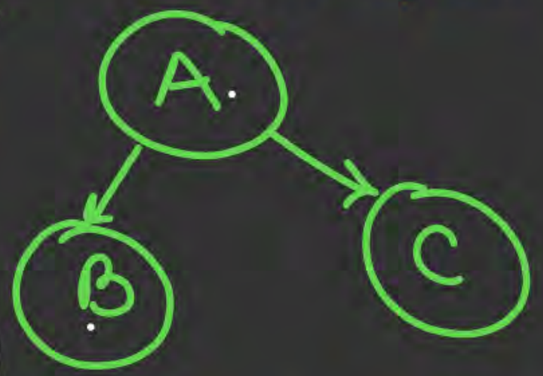
B depend on A
C depend on A

- Simpler structure

- $P(C) \Rightarrow P(C/A)P(A) + P(C/\sim A)P(\sim A)$
  $\Rightarrow P(C, A) + P(C, \sim A)$

- $P(B, C) \Rightarrow \underbrace{P(B/C)}_{\text{B is not dependent on C}} P(C) \Rightarrow P(B)P(C)$

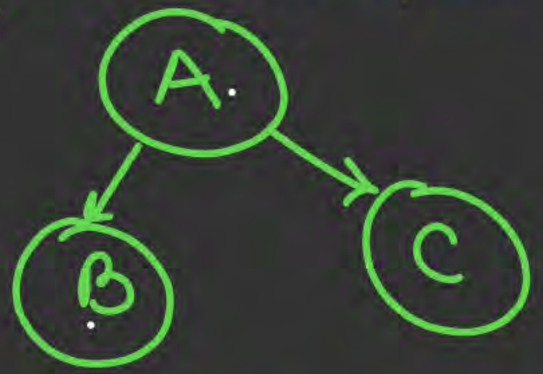
A is indepent of all



B depend on A
C depend on A

- Simplex structure

A is indepent of all



B depend on A
C depend on A

- $P(\sim A, B, C)$

$\Rightarrow P(C/\sim A, B)\, P(\sim A, B)$

$\Downarrow \qquad \Downarrow$

$P(C/\sim A) \cdot P(B/\sim A) \cdot P(\sim A)$

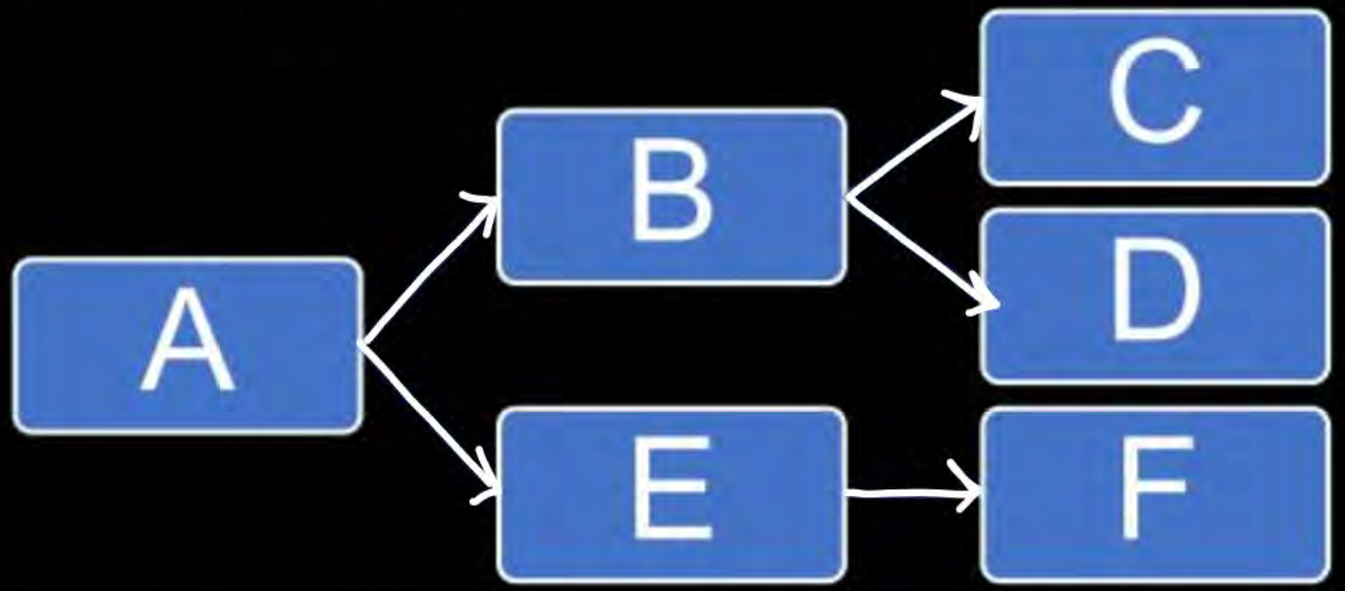
OR

$P(B/\sim A, C)\, P(\sim A, C)$

$\Rightarrow P(B/\sim A)\, P(C/\sim A) \cdot P(\sim A)$

Basics of probability

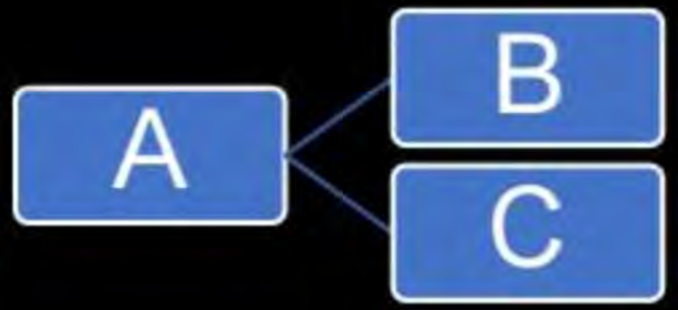
- If we have A,B,C,D,E as random variables
- P(B) = Only depend on A
  $P(B, A) + P(B, \sim A)$
- P(E) = $P(E, A) + P(E, \sim A)$
- P(A) = Fix
- P(C) = $P(C, A, B) + P(C, \sim A, \sim B) + P(C, \sim A, B) + P(C, A, \sim B)$



Basics of probability

- Independent RVs
- P(ABC) = (we always start from bottom)

done



Basics of probability

$P(A=T, B=T, D=T, E=T)$ $\quad P(E/ABD) = P(E/AB)$

$\rightarrow P(E/B)$

- Independent RVs
- P(ABDE) = (we always start from bottom)

$P(E/ABD)\,P(ABD)$

$P(E/B)\,P(D/AB)\,P(AB)$

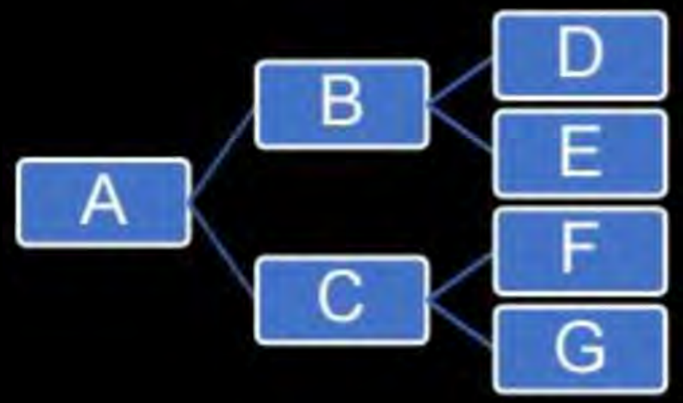
$\Rightarrow \underbrace{P(E/B)\,P(D/B)\,P(B/A)\,P(A)}$

$$P(E) = P(E, B) + P(E, \sim B)$$
$$= P(E, B, \sim A) + P(E, B, A) + P(E, \sim B, A) + P(E, \sim B, \sim A)$$

Basics of probability

- Independent RVs
- P(F) = P(F, C, A)
  + ~C, ~A
  + ~C, A
  + C, ~A
- P(E)

Same here



Basics of probability

- $\Rightarrow P(B, F, A) \Rightarrow P(B, F, A, C) + P(B, F, A, \sim C)$
- $\Rightarrow P(C, G, E) \Rightarrow P(A, B, C, GE) + - - - -$

A, B add 4 Combo

$P(G, D, \sim B, A)$

$\Rightarrow P(G / \cancel{D}, \cancel{\sim B}, A)\, P(D, \sim B, A)$

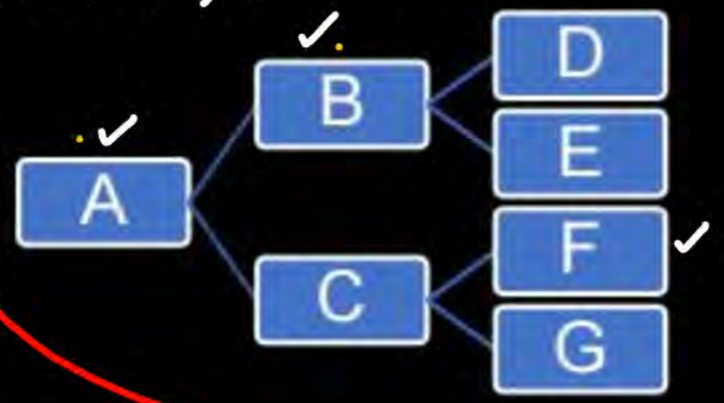
$\Downarrow$

$\Rightarrow P(G|C, A) + P(G|\sim C, A)$

$\Downarrow$

$P(D|\sim B, A)\, P(\sim B, A)$

$\boxed{P(D|\sim B)\, P(\sim B|A)\, P(A)}$



$$\left[\begin{array}{l} P(G/C)\,P(C/A)\,P(A) \\ +P(G/\sim C)\,P(\sim C/A)\,P(A) \end{array}\right] \times P(D/\sim B)\,P(\sim B/A)\,P(A)$$

Probabilistic Reasoning in Artificial Intelligence

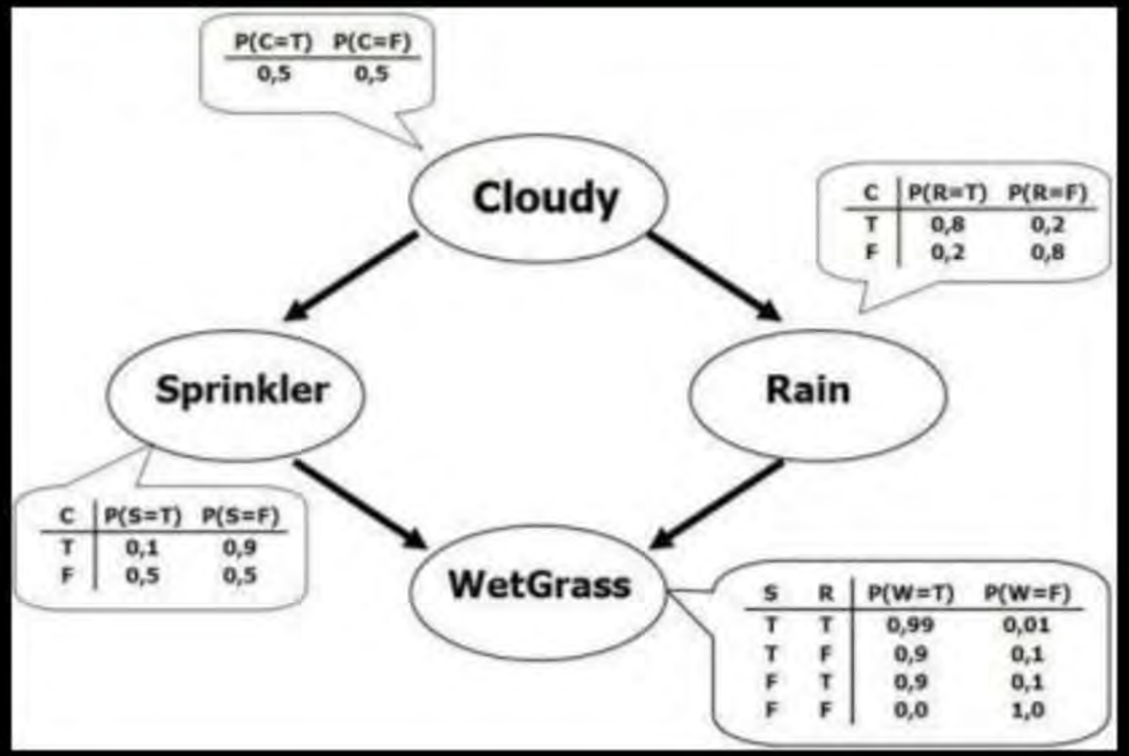
- We have many methods to be used in the probabilistic reasoning
- Bayesian Network
- Markov chain

Bayesian Networks

- A Bayesian Network (BN), also called a Belief Network, is a graphical model that represents probabilistic relationships among a set of variables using a directed acyclic graph (DAG).
- The nodes in the graph represent random variables, and the edges represent conditional dependencies between these variables.
- Each node in a Bayesian Network has a Conditional Probability Distribution (CPD), which defines the probability of the variable given its parents in the network.

Bayesian Networks

- Example
- Cloudy, Sprinkler, Wet Grass, Rain are the Random variable.
- The arrow show the dependencies.



Bayesian Networks

- P(G,S,R)
- P(R/G)

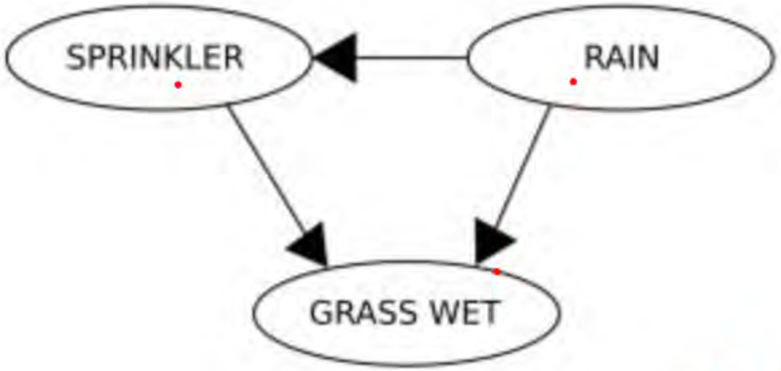
$P(G|SR)P(SR)$

$P(G|SR)P(S|R)P(R)$

$.99 \times .01 \times .2 \Rightarrow 1.98 \times 10^{-3}$

$P(S|\cdot)$

RAIN	SPRINKLER T	SPRINKLER F
F	0.4	0.6
T	0.01	0.99



$P(R)$

RAIN T	RAIN F
0.2	0.8

$P(G|\cdot)$

SPRINKLER	RAIN	GRASS WET T	GRASS WET F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

Bayesian Networks

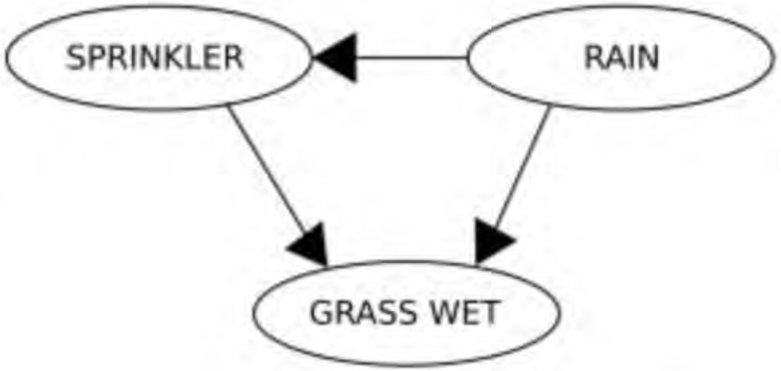
- P(G,S,R)
- P(R/G)

$P(R/G) \Rightarrow \frac{P(R,G)}{P(G)}$

$\Rightarrow P(G) = P(G,R,S) + P(G,\sim R,\sim S) + P(G,\sim R,S) + P(G,R,\sim S)$

$= .99 \times .01 \times .2 + 0 + .9 \times .4 \times .8 + .8 \times .99 \times .2 \Rightarrow .44838$

RAIN	SPRINKLER T	SPRINKLER F
F	0.4	0.6
T	0.01	0.99



RAIN T	RAIN F
0.2	0.8

SPRINKLER	RAIN	GRASS WET T	GRASS WET F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

$\rightarrow P(G,R,S) = P(G|RS)\,P(S|R)\,P(R)$

$P(G,\sim R,\sim S) = P(G|\sim R\sim S)\,P(\sim S|\sim R)\,P(\sim R)$

$P(G,\sim R,S) = P(G|\sim R,S)\,P(S|\sim R)\,P(\sim R)$

$\rightarrow P(G,R,\sim S) = P(G|R,\sim S)\,P(\sim S|R)\,P(R)$

$\Rightarrow P(R,G) \Rightarrow P(RGS) + P(RG\sim S) \Rightarrow .16038$

Ans $\dfrac{P(RG)}{P(G)} = \dfrac{.16038}{.44383}$

Bayesian Networks

- Find P(O,R,~H, C)

$P(O/R{\sim}HC)\,P(R{\sim}HC)$

$P(O/R)\,P(R/{\sim}HC)\,P({\sim}HC)$

$P(O/R)\,P(R|{\sim}HC)\,P({\sim}H)\,P(C)$

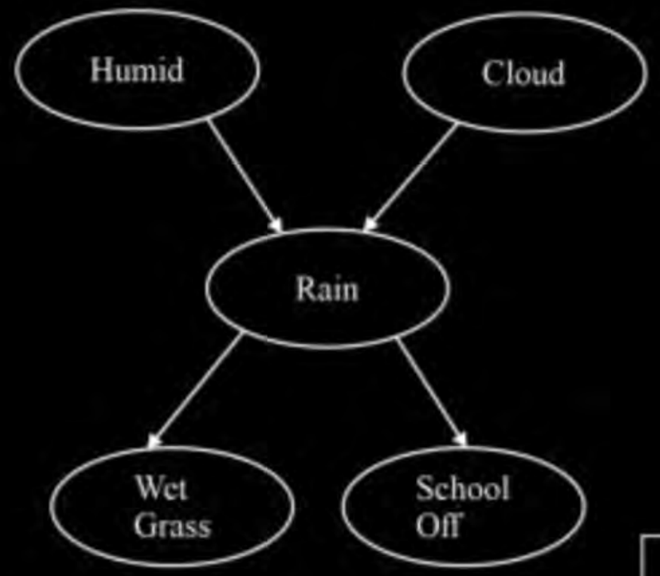
$.7 \times .7 \times .8 \times .6$

$\Rightarrow .2352$

P(H)
0.2

H	C	P(R)
H	C	0.9
H	~C	0.4
~H	C	0.7
~H	~C	0.1

R	P(W)
R	0.8
~R	0.1



P(C)
0.6

R	P(O)
R	0.7
~R	0.2

$P(R|\sim H) \Rightarrow P(R|\sim H \sim C)P(\sim C) + P(R|\sim HC)P(C)$

Bayesian Networks

$P(W O R \sim H)$

$\Downarrow$

$P(W|O R \sim H)P(O R \sim H)$

$P(W|R)P(O|R)\underline{P(R \sim H)}$

$P(W|R)P(O|R)\ P(R|\sim H) \cdot P(\sim H)$

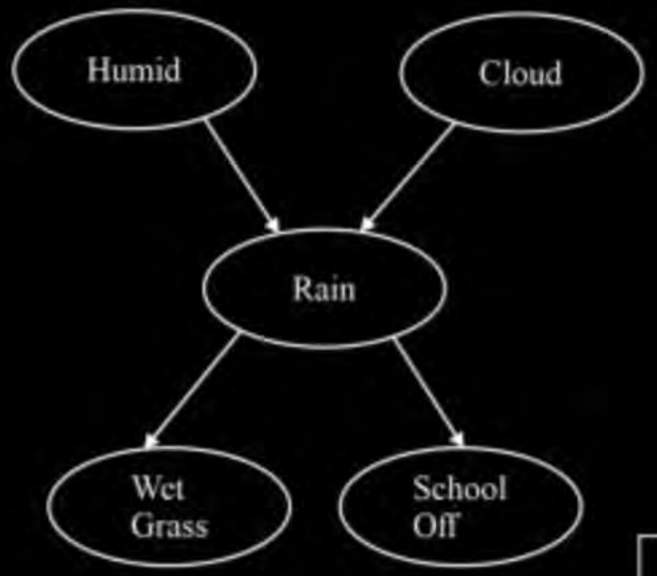
$\cdot 8 \quad \cdot 7 \quad \cdot 46 \quad \cdot 8$

$= (\cdot 20608)$

P(H)
0.2

H	C	P(R)
H	C	0.9
H	~C	0.4
~H	C	0.7
~H	~C	0.1

R	P(W)
R	0.8
~R	0.1



P(C)
0.6

R	P(O)
R	0.7
~R	0.2

$$\boxed{P(R\sim H) = P(R\sim H\sim C) + P(R\sim HC)}$$

$$P(R\sim H) = P(R|\sim H\sim C)P(\sim H\sim C) + P(R|\sim HC)P(\sim HC)$$

$$= P(R|\sim H\sim C)P(\sim H)P(\sim C) + P(R|\sim HC)P(\sim H)P(C)$$

$$P(R|\sim H)P(\sim H) \neq \left(P(R|\sim H\sim C)P(\sim C) + P(R|\sim HC)P(C)\right)P(\sim H)$$

$$P(R|\sim H) \neq P(R|\sim H\sim C) + P(R|\sim HC)$$

Bayesian Networks

- Find P(W/S) $= \dfrac{P(W,S)}{P(S)}$

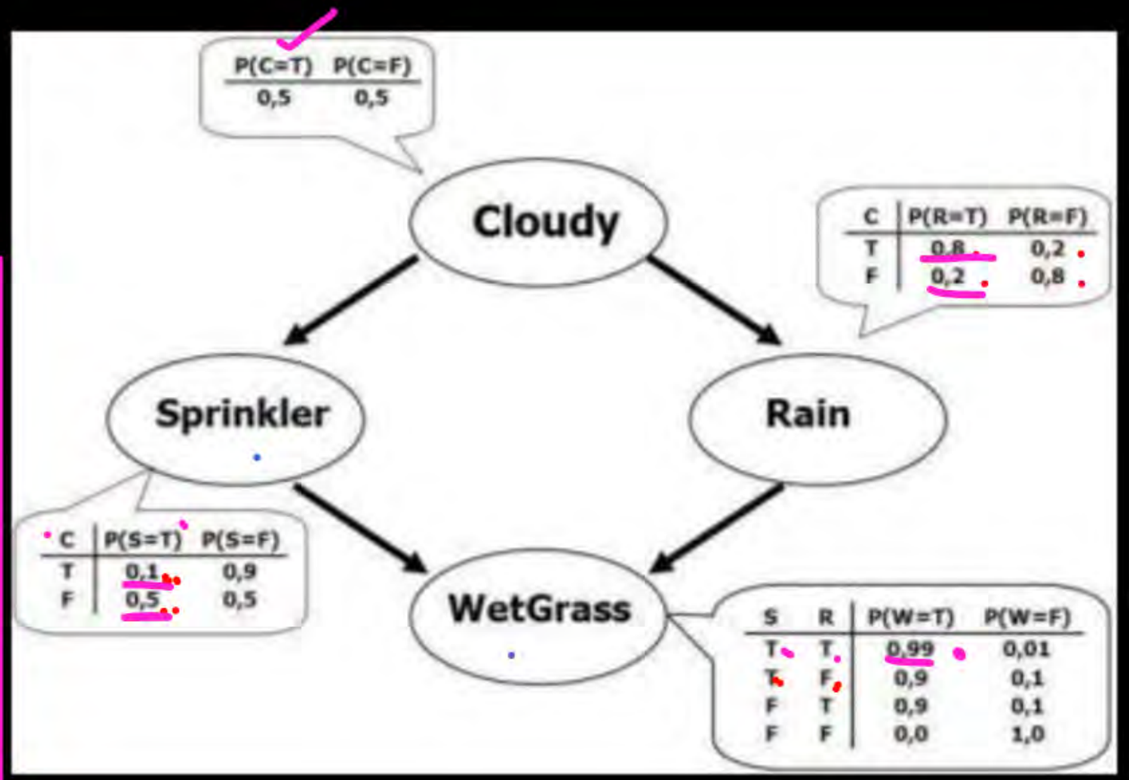
$P(W,S) =$ $P(W,S,R,C) = .99 \times .1 \times .8 \times .5$

R, C Both $+ P(W,S,R,\sim C) \Rightarrow .99 \times .5 \times .2 \times .5$

$+ P(W,S,\sim R,C) \Rightarrow .9 \times .1 \times .2 \times .5$

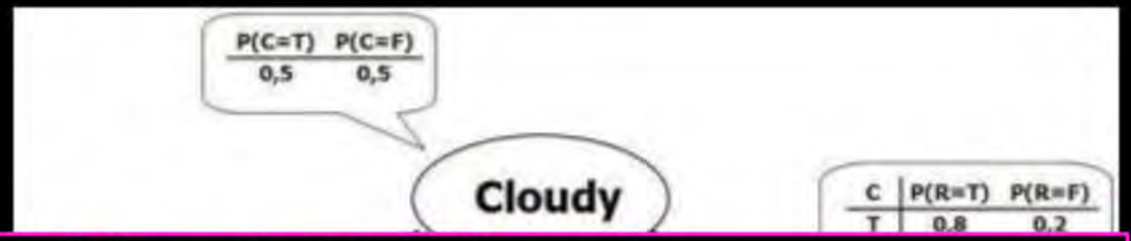
$+ P(W,S,\sim R,\sim C) \Rightarrow .9 \times .5 \times .8 \times .5$

$\Rightarrow .2781$



Bayesian Networks

- Find P(W/S) $= \dfrac{P(W,S)}{P(S)}$



$P(W,S) = P(W,S,R,C) \Rightarrow P(W/SRC) \cdot P(S/CR) \Rightarrow P(W|SR)\, P(S|RC)\, P(RC)$

$\Rightarrow P(W|SR)\, P(S|C)\, P(R|C)\, P(C)$

R, C Both $+ P(W,S,R,\sim C) \Rightarrow P(W/R,S)\, P(S/\sim C)\, P(R/\sim C)\, P(\sim C)$

$+ P(W,S,\sim R,C) \Rightarrow P(W/\sim R,S)\, P(S|C)\, P(\sim R|C)\, P(C)$

$+ P(W,S,\sim R,\sim C) \Rightarrow P(W|S\sim R)\, P(S/\sim C)\, P(\sim R/\sim C)\, P(\sim C)$

Bayesian Networks

$P(s) = P(s|c)P(c) + P(s|\sim c)P(\sim c)$

$\boxed{P(c|s) \neq P(c)}$

By Bayes theorem ⇒

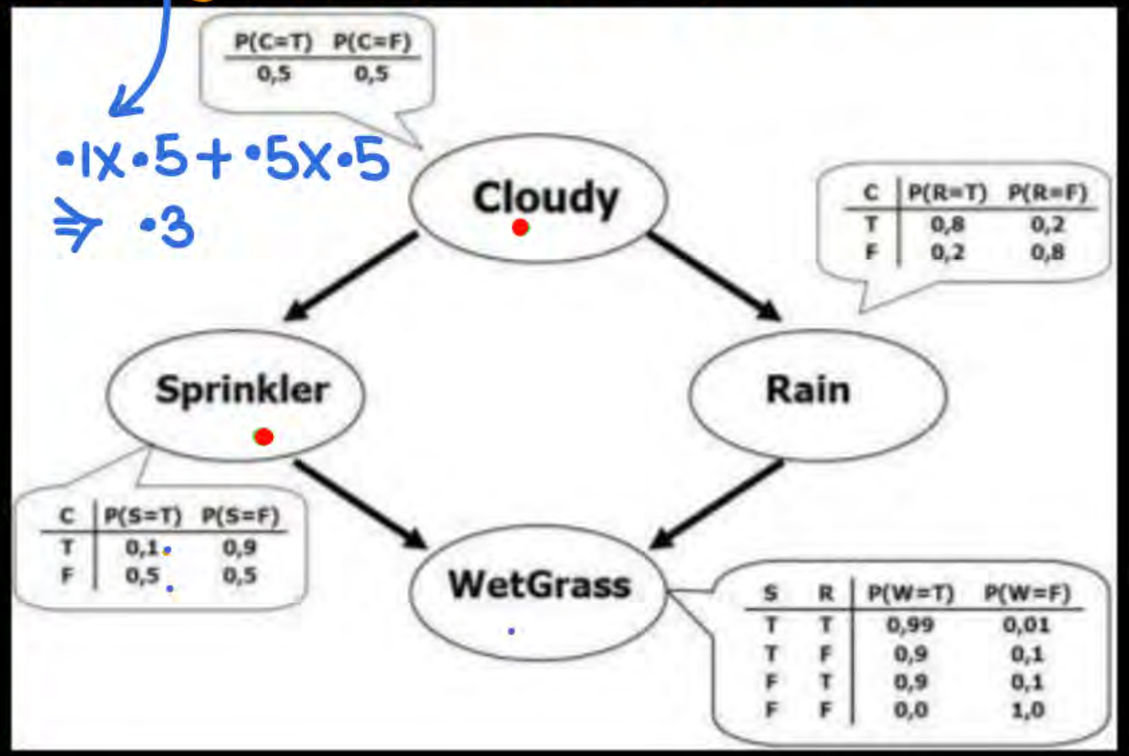
$P(c|s)P(s) = P(s|c)P(c)$

$P(c|s) = \dfrac{P(s|c)P(c)}{P(s)} \Rightarrow \dfrac{.1 \times P(c)}{.3}$

$P(c|s) \Rightarrow .5 \times .1/.3 \Rightarrow (.166)$

$.1 \times .5 + .5 \times .5$

$\Rightarrow .3$



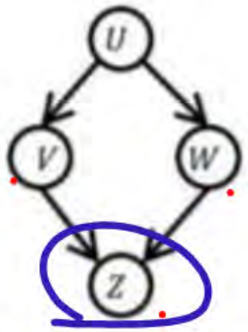
P(C=T)	P(C=F)
0,5	0,5

C	P(R=T)	P(R=F)
T	0,8	0,2
F	0,2	0,8

C	P(S=T)	P(S=F)
T	0,1	0,9
F	0,5	0,5

S	R	P(W=T)	P(W=F)
T	T	0,99	0,01
T	F	0,9	0,1
F	T	0,9	0,1
F	F	0,0	1,0

Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \mid \cdot)$	$P(V = 1 \mid \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \mid \cdot)$	$P(W = 1 \mid \cdot)$
$U = 0$	1	0
$U = 1$	0	1

		$P(Z = 0 \mid \cdot)$	$P(Z = 1 \mid \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) =$ .125 (rounded off to **three** decimal places).

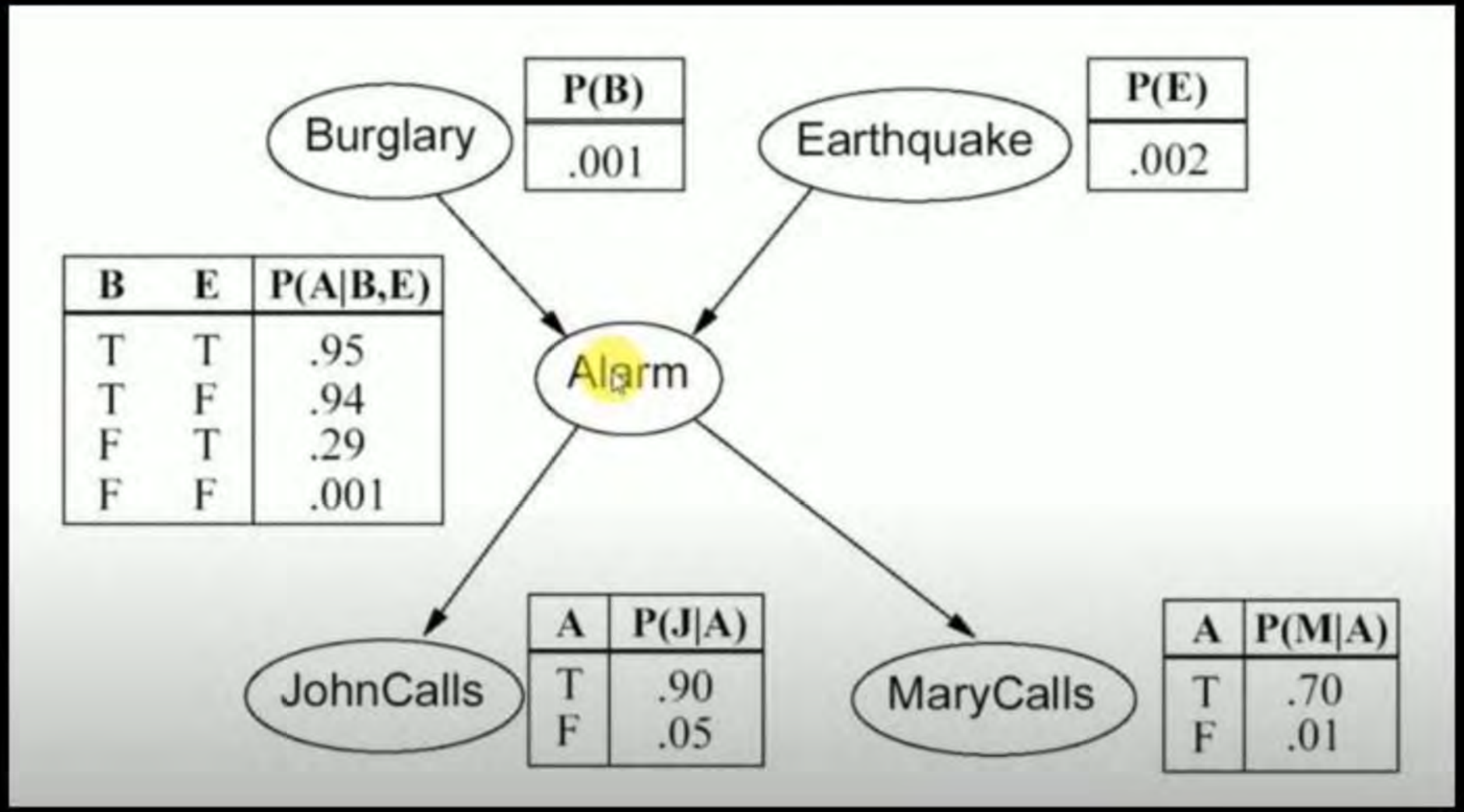
$P(u,v,w,z) \Rightarrow P(z/uvw)P(uvw)$

$\Rightarrow P(z/vw)\, P(v/uw)\, P(uw)$

$\Rightarrow P(z/vw)P(v/u)P(w/u)P(u)$

$\Rightarrow .5 \times .5 \times 1 \times .5 \Rightarrow (.125)$

Bayesian Networks



P(B)
.001

P(E)
.002

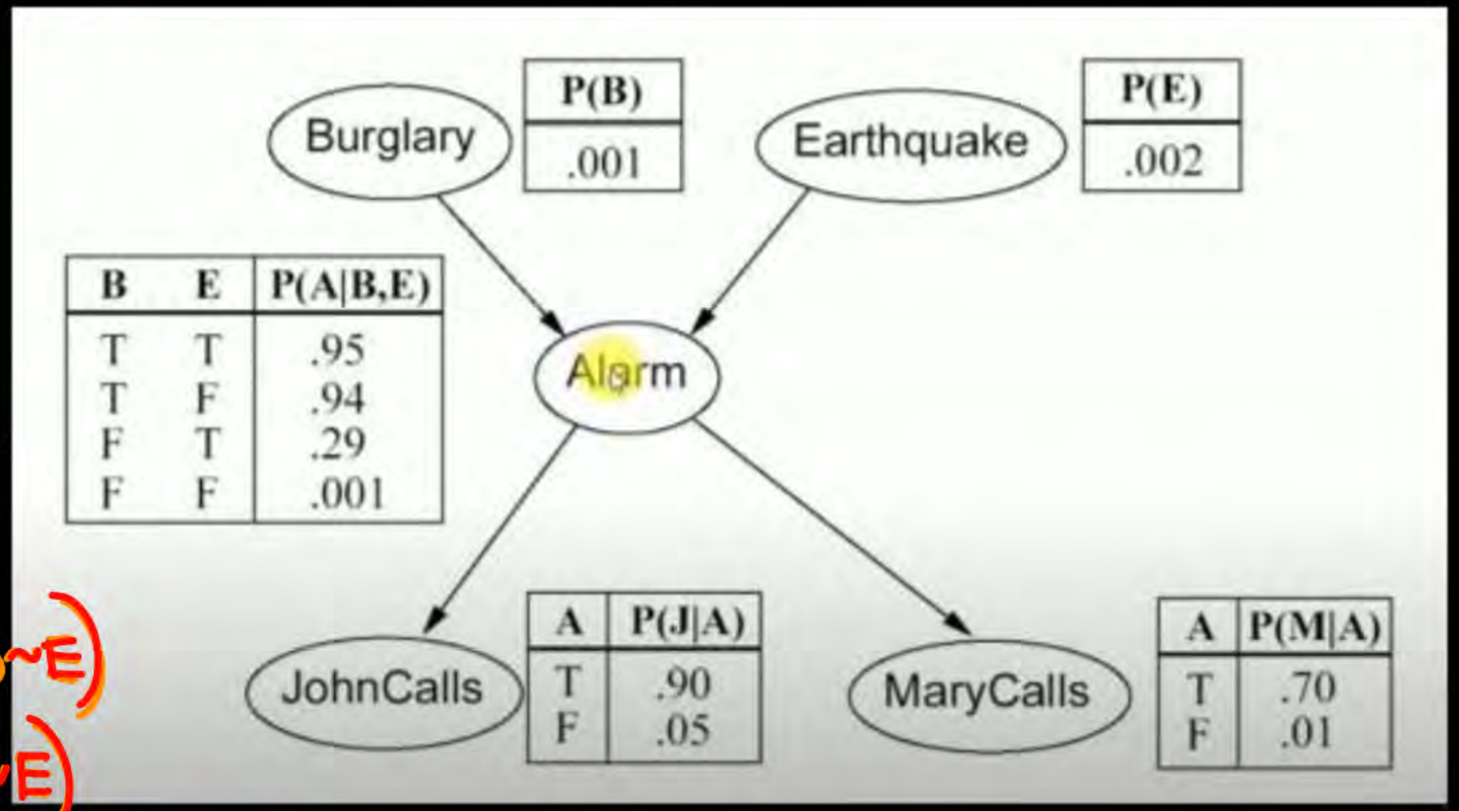
B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

Bayesian Networks

- What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



$P(A \sim B \sim E J M)$

$\Rightarrow P(J/A \sim B \sim E M)\, P(M A \sim B \sim E)$

$\Rightarrow P(J/A)\, P(M/A)\, P(A \sim B \sim E)$

$\Rightarrow P(J/A)\, P(M/A)\, P(A/\sim B \sim E)\, P(\sim B \sim E)$

$$P(J/A)\,P(M/A)\,P(A/\sim B \sim E)\,P(\sim B)\,P(\sim E)$$

$$.9 \times .7 \times .001 \times .999 \times .998$$

$$.000628$$

Bayesian Networks

2. What is the probability that John call?

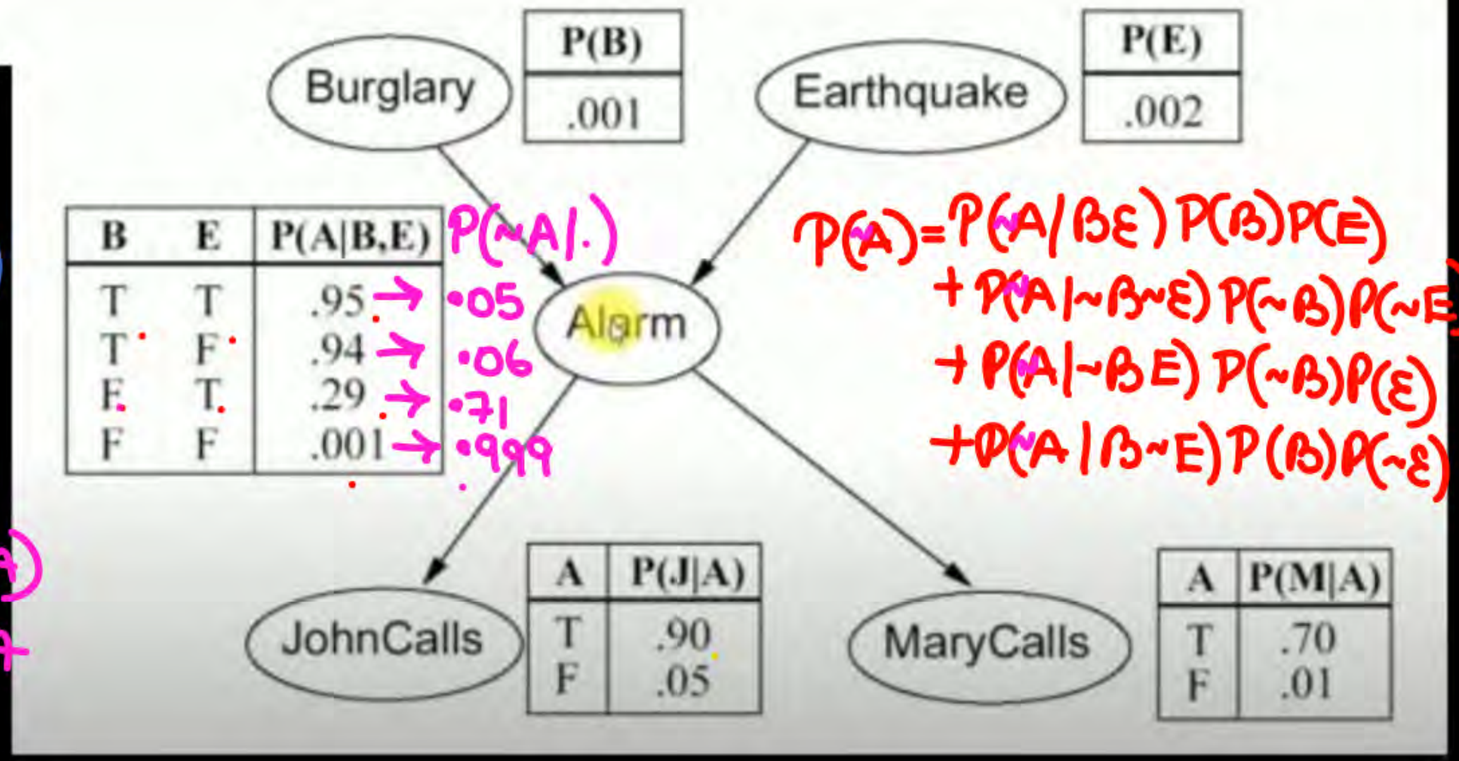
$P(J) \Rightarrow P(J, A) + P(J, \sim A)$

$\Rightarrow P(J/A)P(A) + P(J/\sim A)P(\sim A)$

(2.51×10^{-3})

$P(\sim A) = 1 - P(A)$
$= .997$

$\Rightarrow .052109$



P(B)
.001

P(E)
.002

B	E	P(A|B,E)	P(~A|.)
T	T	.95 →	.05
T	F	.94 →	.06
F	T	.29 →	.71
F	F	.001 →	.999

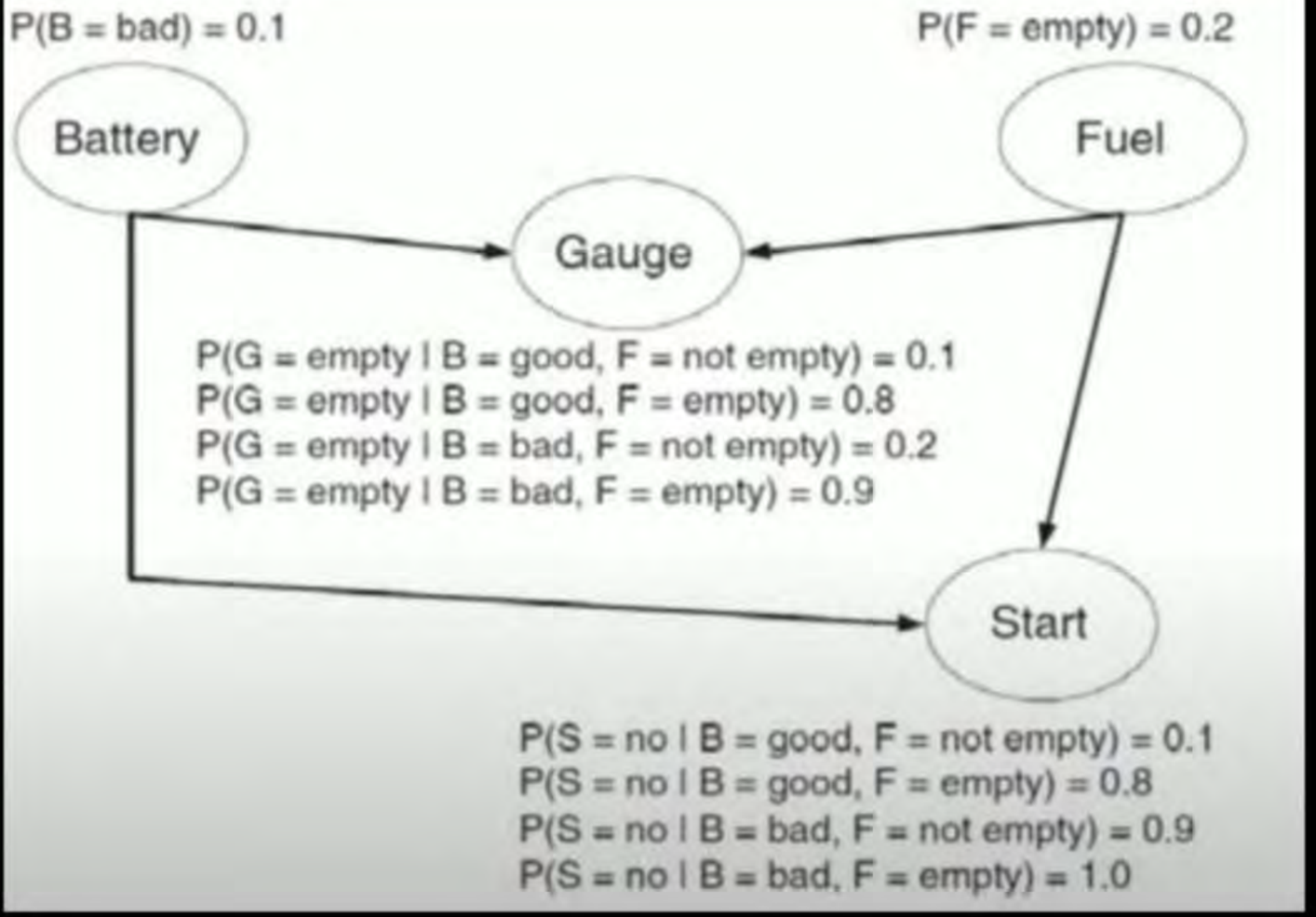
A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

$P(A) = P(A/BE)P(B)P(E)$
$+ P(A/\sim B\sim E)P(\sim B)P(\sim E)$
$+ P(A/\sim BE)P(\sim B)P(E)$
$+ P(A/B\sim E)P(B)P(\sim E)$

Bayesian Networks

1. P(B=Good, F=Empty, G=Empty, S=Yes) H.W
2. P(B=Bad, F=Empty, G=Not Empty, S=No)
3. Given the battery is bad, Compute the probability that the car will start



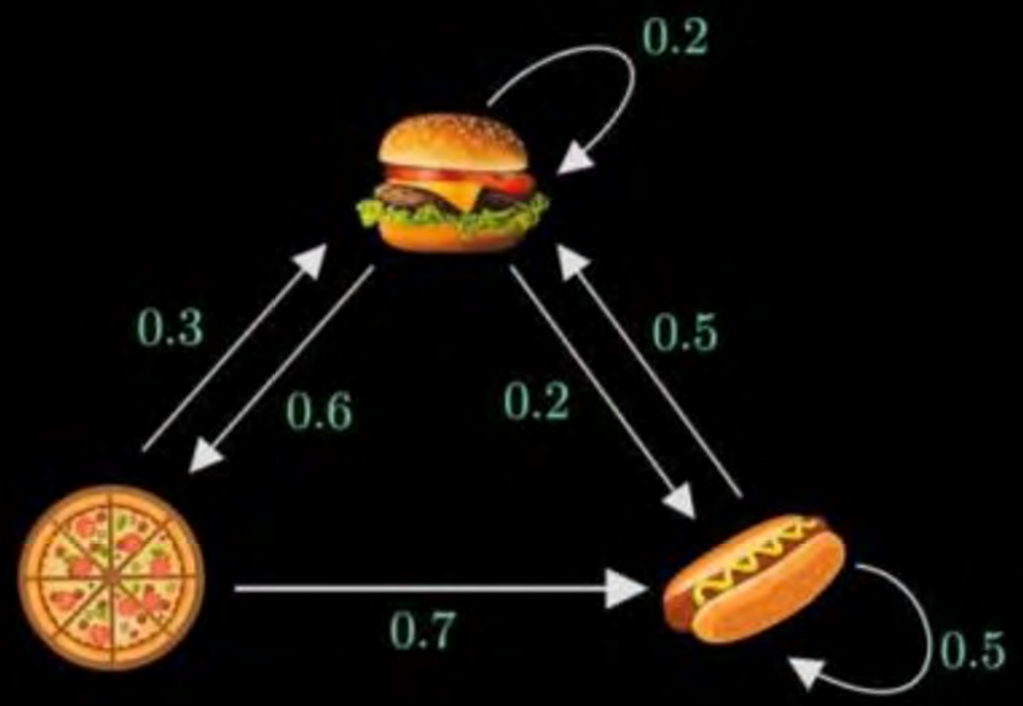
Bayesian Networks

- Components of a Bayesian Network
- Nodes: Each node represents a random variable. Variables can be binary, discrete, or continuous.
- Edges: Directed edges between nodes represent dependencies. If there's an edge from node A to node B, it means that B depends probabilistically on A.
- Conditional Probability Tables (CPTs): For each node, a CPT defines the probability of the node's possible states given the states of its parent nodes.

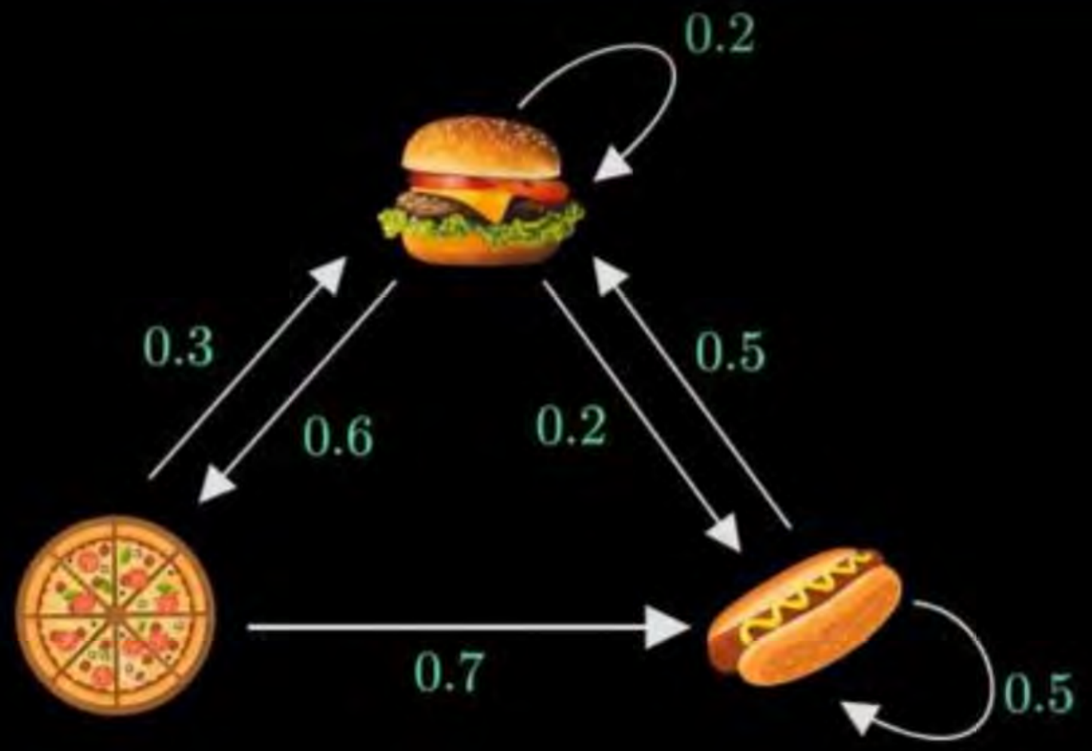
Markov Chain

- A Markov Chain is a mathematical system that undergoes transitions from one state to another within a finite or
- countable number of possible states. It is a type of stochastic process that satisfies the Markov property, which states
- that the future state depends only on the current state and not on the sequence of events that preceded it.

Markov Chain



Probability that hot dog will be served 4th day…



$$P(X_{n+1} = x \mid X_n = x_n)$$

Markov Chain

- Key Concepts: State Space (S):A set of all possible states that a system can be in. For example, in a weather prediction model, the states could be {Sunny, Rainy, Cloudy}.
- Transition Probabilities (P):The probability of moving from one state to another. It is often represented as a matrix P, where P ij represents the probability of moving from state i to state j. The sum of probabilities for transitions from any state must equal 1.

Markov Chain

- Initial State Distribution (π):A probability distribution over the initial states of the system, representing the probability of starting in any given state.
- Transition Diagram:A graphical representation of the states and the transition probabilities between them. Nodes represent states, and directed edges represent the probabilities of moving from one state to another.

Markov Chain

- Example: Weather Prediction Scenario:
- We are modeling the weather using two states: State 1 (Sunny) State 2 (Rainy)Transition Matrix: Suppose we are given the following transition matrix P:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

- $P_{11} = 0.7$: Probability of remaining sunny.
- $P_{12} = 0.3$: Probability of going from sunny to rainy.
- $P_{21} = 0.4$: Probability of going from rainy to sunny.
- $P_{22} = 0.6$: Probability of remaining rainy.

Markov Chain

4. Markov Property

The defining property of a Markov chain is its **memorylessness**, mathematically expressed as:

$$P(X_{n+1} = s_{n+1} | X_n = s_n, X_{n-1} = s_{n-1}, \ldots, X_0 = s_0) = P(X_{n+1} = s_{n+1} | X_n = s_n)$$

This implies that the future state depends only on the present state, not on the sequence of events that preceded it.

Markov Chain

- **Stationary Distribution:**
  A stationary distribution is a probability distribution over the states that does not change as the system evolves. It satisfies the equation:

$$\pi P = \pi$$

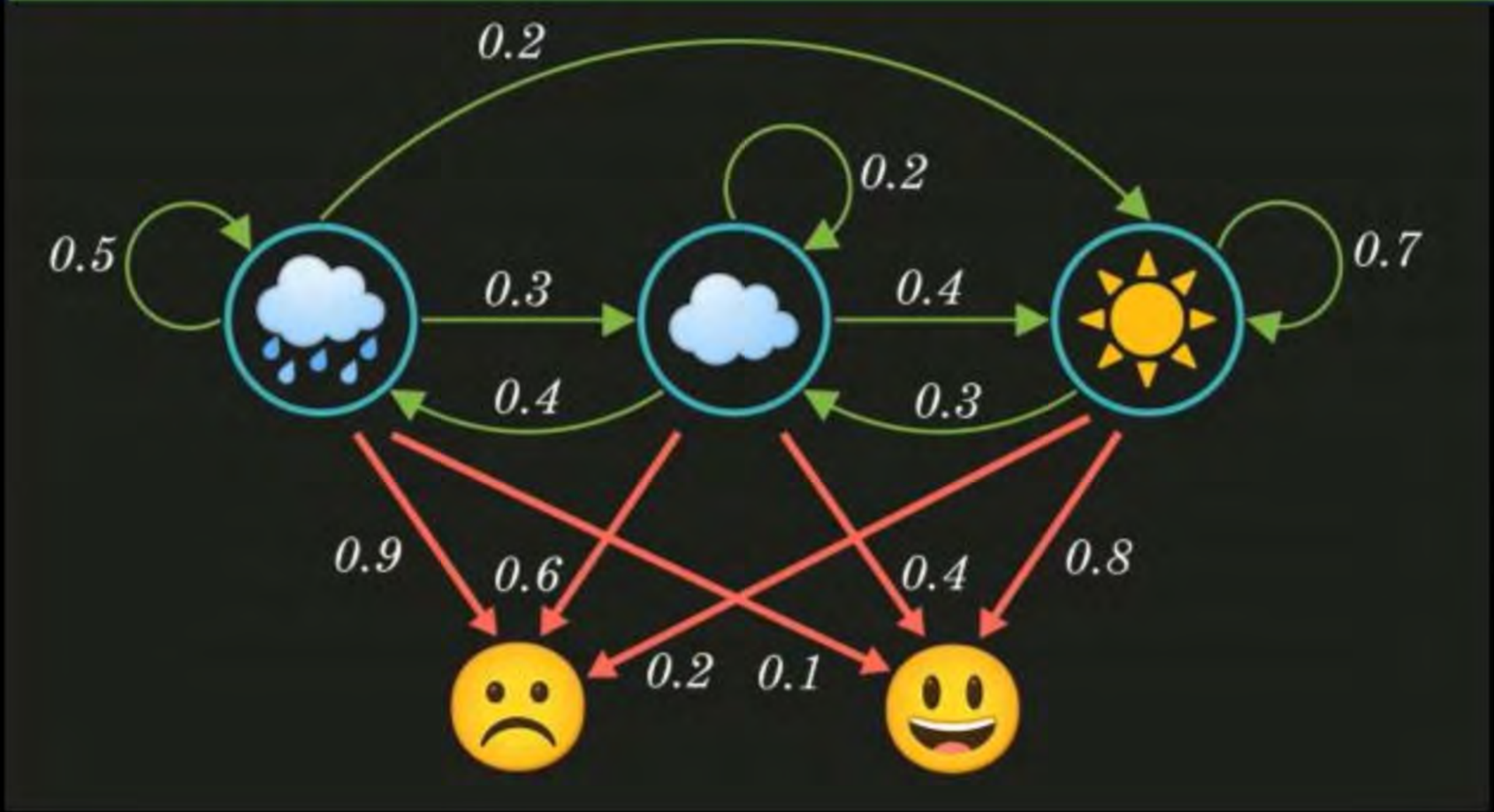
  This means that if the system starts in the stationary distribution, it will remain in that distribution after each transition.

Hidden Markov Model (HMM) Overview

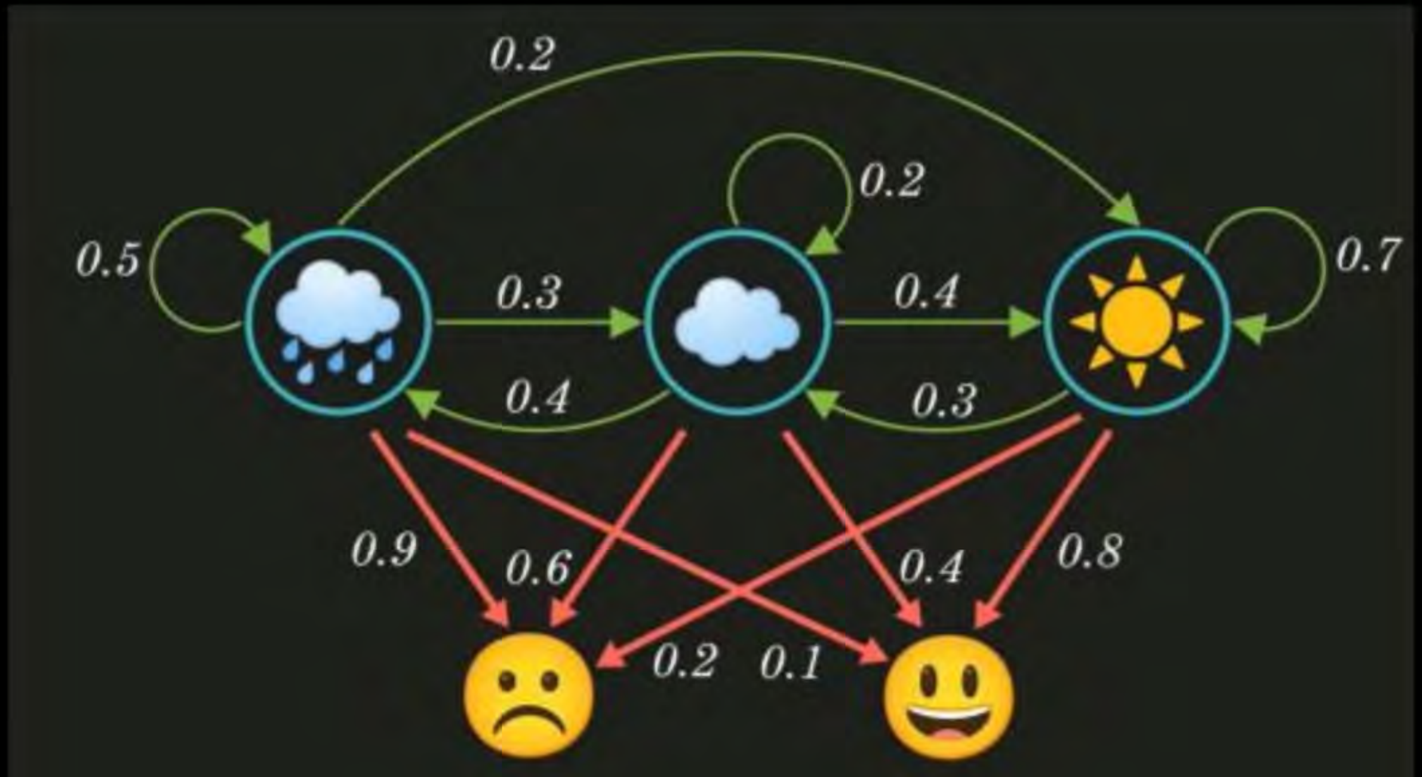
- Hidden Markov Models (HMMs) are statistical models used to describe systems that evolve over time. They are particularly useful in cases where the system is "hidden" (not directly observable), but you can observe some output that provides information about the underlying system.
- A Hidden Markov Model (HMM) is an extension of a Markov Chain where the states are not directly observable (hidden). Instead, each state produces an observable output (or emission) according to a probability distribution

Hidden Markov Model (HMM) Overview – the simple markov chain

So we don't live in the city of my friend hence we dont know the weather of the city hence they are not known to us or hidden to us, but we know the mood of the person



The present day mood of my friend depend only on the present day weather and not on previous day mood...



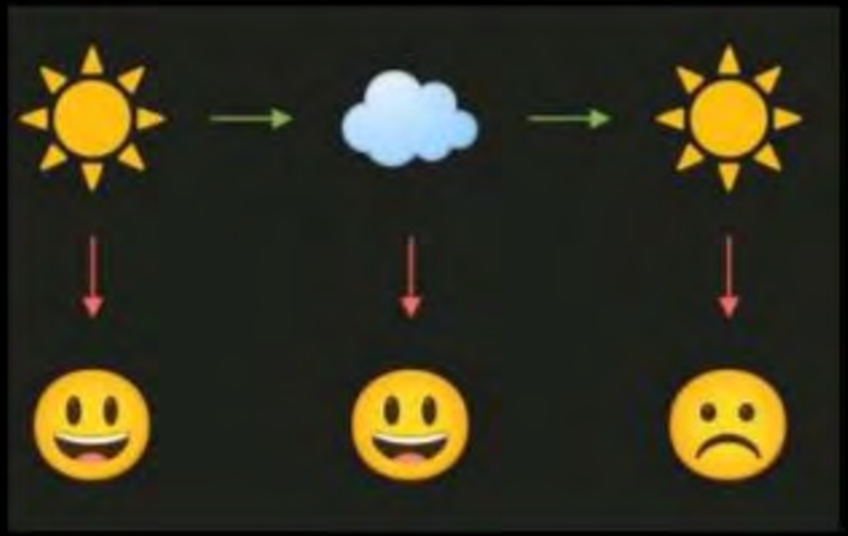
HMM = Hidden MC + Observed Variables

The transition matrix and the emission matrix

HMM = Hidden MC + Observed Variables

$$\begin{array}{c|ccc} & \text{🌧} & \text{☁} & \text{☀} \\ \hline \text{🌧} & 0.5 & 0.3 & 0.2 \\ \text{☁} & 0.4 & 0.2 & 0.4 \\ \text{☀} & 0.0 & 0.3 & 0.7 \end{array} \qquad \begin{array}{cc} \text{☹} & \text{😃} \\ \hline 0.9 & 0.1 \\ 0.6 & 0.4 \\ 0.2 & 0.8 \end{array}$$

Find the probability of this sequence ...



THANK - YOU